

Refly.ai Parity Report

Integrations, features, current limitations, and proposed work

Overview

This document summarizes the target feature parity with Refly.ai, our current implementation status in the Angular + LangGraph app, identified limitations, and a prioritized roadmap.

- Angular
- LangGraph
- OpenRouter
- Postgres + pgvector
- MinIO (S3)

Integrations

- **Model provider:** OpenRouter (multi-model access; chat + embeddings).
- **LangGraph:** Journeys/executor for multi-step flows.
- **Database:** Postgres with **pgvector** for semantic search.
- **Object storage:** MinIO (S3-compatible) for file uploads.
- **Frontend:** Angular app with routes for Chat, Canvas, Editor.
- **Observability:** Request logging (server); LangSmith integration planned.

Feature Parity

Delivered

- Canvas module: nodes, SVG edges, dragging.
- AI canvas generator via LLM (strict JSON parsing).
- Knowledge panel: upload (S3), URL ingest, semantic search, citation copy.
- Editor: AI summarize/rewrite, knowledge search, citation insert, MD/HTML export.
- Chat: multi-thread with persistence; workspace switcher.
- Journeys: minimal executor using LangGraph.

Planned / In Progress

- Auth + RBAC per workspace (roles, membership).
- Rich editor (blocks, templates, citations manager, PDF export).
- Knowledge graph relations and advanced filtering.
- Observability via LangSmith (traces, runs, datasets).
- Code/diagram generation: Mermaid live preview, HTML/SVG widgets.
- Collaboration (CRDT) for real-time canvas/editor.

- Workspaces API with DB fallback; basic request logging.

Current Limitations

- **DB optional mode:** When Postgres is down, some endpoints fall back to defaults; persistence is limited.
- **Editor:** Basic text area (no full block editor, no bibliography manager yet).
- **Canvas:** Edges render straight lines; no pan/zoom or connector handles yet.
- **Search:** No hybrid BM25 + vector; re-ranking not yet implemented.
- **Auth:** No user identities or access control (coming with RBAC).
- **Observability:** Tracing to LangSmith not wired; logs are server console only.

Proposed Work (Roadmap)

1. **Auth & Workspaces:** Add OAuth, user profiles, memberships, roles. Lock resources to workspace.
2. **Observability:** Add LangSmith tracing to chat, journeys, and ingestion; audit logs.
3. **Editor Upgrade:** Move to block editor (ProseMirror/Tiptap), citation manager, PDF export.
4. **Canvas UX:** Pan/zoom, edge routing, node types (website, memo, code), templates UI.
5. **Knowledge:** Hybrid retrieval (BM25 + vector), re-ranking, knowledge graph links.
6. **Agents:** Specialized LangGraph agents (researcher, writer, reviewer) with supervisor.
7. **Collaboration:** Yjs-based CRDT for real-time co-editing on canvas/editor.
8. **Packaging:** Docker/K8s manifests and environment hardening.